Estimation of Rooftop Solar Photovoltaic Potential based on Deep Learning

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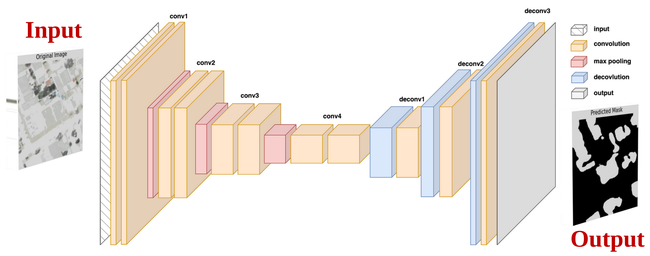
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1. Introduction

In this project, Machine learning techniques were utilized to determine the suitability of rooftops in New York for solar panel installation, thereby promoting environmental sustainability. Specifically, a semantic segmentation model was trained using RGB and Near-Infrared (NIR) data with a U-Net architecture to create a mask that identifies rooftop areas. This mask serves as a foundational step in identifying rooftops with significant potential for renewable energy production. The U-Net model, known for its effectiveness in semantic segmentation tasks, demonstrated high accuracy in distinguishing between rooftops and other urban features, ensuring precise identification of suitable areas.

2. Methodology

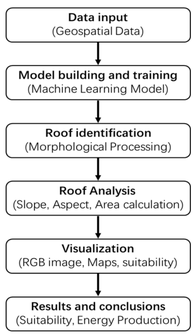
2.1. Model Building&Training



1.Each encoder and decoder block contains:
- Two consecutive **3×3** convolutional layers
- BatchNorm2d after each convolution
- ReLU activation function **(inplace = True)**

2.Skip Connection Implementation:
- Feature concatenation using torch.cat
- Decoder input channels = upsampled channels + corresponding encoder output channels

3.Training Configuration:
- Optimizer: Adam (learning_rate =**1e-5**)
- Loss Function: Binary Cross Entropy (**BCELoss**)
- Batch Size: **8**
- Early Stopping: patience =**10**



2.2. Solar Photovoltaic Potential Estimation

1.Roof area: $A_{roof} = N_{pixels} \times A_{pixel}$

2.Roof Characteristics

1.Slope Calculation:

$$\text{slope} = \arctan\left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\right) \times \frac{180}{\pi}$$

2.Aspect (Direction)Calculation:

$$\text{aspect} = \left(\arctan 2\left(\frac{\partial z}{\partial y}, \frac{\partial z}{\partial x}\right) \times \frac{180}{\pi}\right) \bmod 360$$

3.Suitability Score:

1.Slope Score:

$$S_{slope} = \begin{cases} 1.0 & \text{if } 20^\circ \le \text{slope} \le 40^\circ \\ 0.7 & \text{if } 10^\circ \le \text{slope} \le 50^\circ \\ 0.3 & \text{otherwise} \end{cases}$$

2.Aspect Score:

$$S_{aspect} = \cos(\text{aspect} - 180^\circ) \times 0.5 + 0.5$$

3.Thermal Score: $S_{thermal} = \frac{\text{thermal}_{\text{roof}}}{\text{thermal}_{\max}}$

4.Final Suitability score:

$$S_{final} = \begin{cases} 0.9 & \text{if flat roof} \\ 0.4S_{slope} + 0.4S_{aspect} + 0.2S_{thermal} & \text{otherwise} \end{cases}$$

5. Solar Panel installation

1.Panel Area:

$A_{\text{panel}} = (W_{\text{panel}} + S_{\text{spacing}}) \times (H_{\text{panel}} + S_{\text{spacing}})$

- $W_{\text{panel}} = 1.70\,\text{m}, H_{\text{panel}} = 1.00\,\text{m}$: Panel dimensions.
- $S_{\text{spacing}} = 0.05\,\text{m}$: Spacing between panels.

2. Daily Energy (kWh):

$$E_{\text{daily}} = \frac{N_{\text{panels}} \times P_{\text{panel}} \times H_{\text{sun}} \times S_{\text{final}} \times R_{\text{performance}}}{1000}$$

- $P_{\text{panel}} = 400\,\text{W}$: Power of a single panel.
- $H_{\text{sun}} = 5.0\,\text{hours}$: Daily sunlight hours.
- $R_{\text{performance}} = 0.75$: Performance ratio.

3. Annual Energy Production:

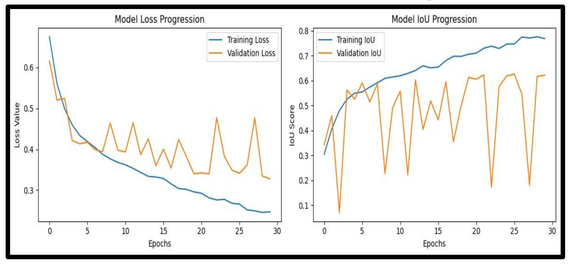
$$E_{annual} = E_{daily} \times 365$$

6. Annual Economic Benefit:

$$B_{annual} = E_{annual} \times 0.5\,\text{USD/kWh}$$

3. Result & Discussion

3.1. Result of Model Testing:



Iou:0.6139, Dice:0.7593, Precision:0.7852, Recall:0.7370, F1:0.7593

Model Loss Progression:
- Illustrates changes in training and validation loss during training.
- **Training loss:** Gradually decreases as training progresses.
- **Validation loss:** Fluctuates more significantly, indicating potential instability in model performance on the validation set.

Model IoU Progression:
- Displays Intersection over Union (IoU) scores for both training and validation sets.
- **Training IoU:** Steadily increases over time.
- **Validation IoU:** Shows greater variability, potentially due to data complexity or model overfitting.

3.2. Result of Model Prediction:

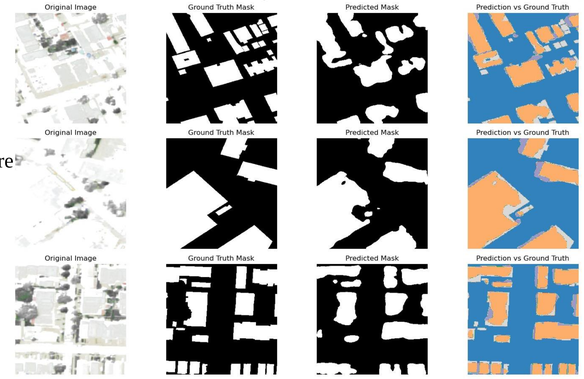
Challenges Identified:

1.Edge Definition:
Predicted masks are smoothed and rounded, missing sharp building edges due to challenges like shadows or obstructions.

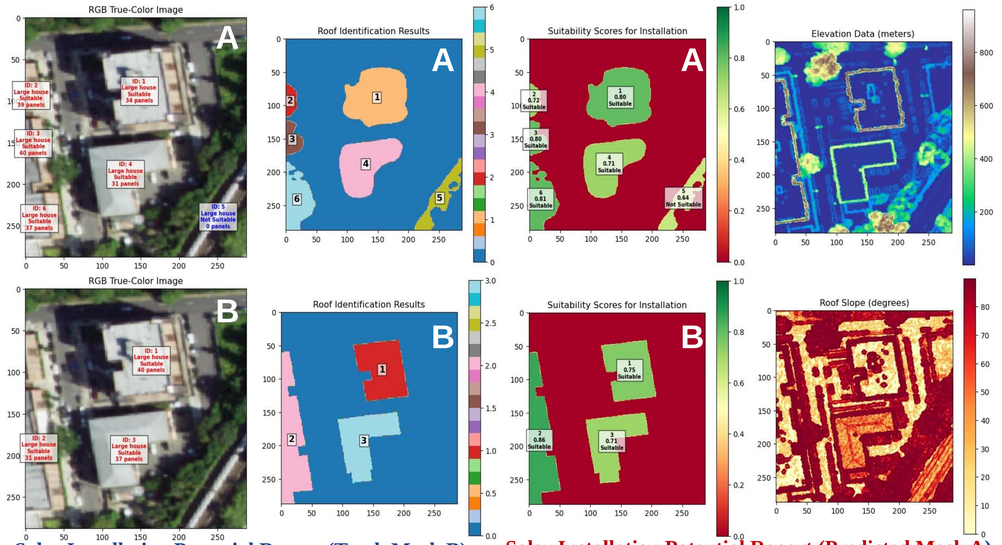
2.Detail Loss:
Small structures and intricate building outlines are simplified, reducing accuracy for solar planning.

3.Building Separation:
Closely spaced buildings are often merged, hindering per-building analysis.

4.Scale Bias:
Larger buildings are captured better, while smaller structures show inconsistent segmentation.



3.3. Solar Photovoltaic Potential Prediction Comparison:



Solar Installation Potential Report (Truth Mask B)
Number of identified roofs: **3**
Number of suitable roofs: **3**
Total number of solar panels that can be installed: **108**
Estimated annual energy production: **45490.1 kWh**
Estimated annual economic benefit: **22745.0 USD**

Solar Installation Potential Report (Predicted Mask A)
Number of identified roofs: **6**
Number of suitable roofs: **5**
Total number of solar panels that can be installed: **181**
Estimated annual energy production: **76207.7 kWh**
Estimated annual economic benefit: **38103.8 USD**

4. Conclusion

View the test set evaluation results:
- **IoU: 0.6139** - indicates that the model prediction overlaps with the real building outline by 61.39%, which is a good result
- **Dice coefficient: 0.7593** - another indicator of segmentation quality, a score close to 0.76 indicates an overall good segmentation
- **Precision: 0.7852** - indicates that 78.52% of the areas predicted by the model to be buildings are indeed buildings
- **Recall: 0.7370** - indicates that the model successfully recognized 73.70% of all real building areas
- **F1 score: 0.7593** - the harmonic mean of precision and recall, reflecting the overall balance of the model

From the visualization results:
- The model is able to capture the main contours of buildings well, especially on large buildings
- There is still room for improvement in the recognition of some details, such as building edges and small buildings
- In the comparison chart (rightmost column), the blue area indicates correct predictions, and the orange area shows the part of the model's misjudgment

5. Future Improvement

Dataset Scale Optimization
Expand from the current 1,000 samples to the complete 30,000-image dataset in phases **(5,000, 10,000, 30,000).** Increase training epochs from **30 to 80-100**, coupled with dynamic learning rate adjustment to ensure thorough feature learning by the model.

Data Diversity Enhancement
Supplement with seasonal data (changes across four seasons), meteorological data (sunlight hours, cloud coverage), and higher resolution imagery. This additional information will enhance the model's understanding of building characteristics and environmental conditions, improving prediction accuracy.

Model Architecture Improvement
Enhance the existing U-Net model by incorporating attention mechanisms, utilizing pre-trained backbone networks, implementing multi-scale feature fusion.

Assessment System Enhancement
Expand evaluation dimensions to include building age, roof materials, and environmental factors. Establish a machine learning-based weight adaptation system that optimizes factor weightings based on real-world cases, ensuring assessment results better align with practical situations.

6. Key Reference

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